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Barriers and facilitators to uptake of lung cancer screening: protocol of a mixed methods systematic review

Yu-an Lin[1], Yu ting Hong[1],Bo ni Chen[1], Hui ming Xiao[1] , Fei fei Huang[1]

1. School of Nursing, Fujian Medical University, Fuzhou, China

Email：

Yu-an Lin: lya980211@163.com

Yu ting Hong: ann_hyt@163.com

Bo ni Chen: chenboni2005@163.com

Hui ming Xiao: huimin_xiao@126.com

Correspondence to Fei fei Huang

hff315@fjmu.edu.cn

No 1 Xueyu Road, Minhou county, Fuzhou 350108, Fujian, China.

Telephone number +86 13763869099

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Abstract

Introduction Globally, uptake rates of LCS with low-dose computed tomography remain low. Numerous factors contribute to the underuse of LCS, and theory-informed approaches to identify and address uptake of LCS barriers and facilitators are needed. This study aims to document the methods which will be used to identify, appraise and synthesize the available qualitative, quantitative, and mixed-methods evidence to address the barriers and facilitators at the individuals and healthcare providers level according to the social-ecological model, and identify gaps to inform practice and policy.

Methods and analysis The following databases will be searched: PUBMED, Ovid (Journals @ Ovid Full Text and Ovid MEDLINE), EMBASE, CINAHL, PsycINFO, Cochrane Library, Chinese Biomedical Database (CBM), Chinese National Knowledge Infrastructure (CNKI), Wanfang database, from inception to December 31, 2020. Two reviewers will be involved in independently screening, reviewing and synthesizing the data and calibration exercises will be conducted at each stage. Disagreements between the two authors will be resolved by arbitration by a third author. The NIH Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies, the Critical Appraisal Skills Programme criteria adapted for qualitative studies and the 16-item quality assessment tool (QATSDD) are used in Quality assessment of primary studies. We will perform data synthesis using the Review Manager (RevMan) software V.5.3.

Ethics and dissemination This study will be a review of published data and it's not necessary to obtain ethical approval. Findings of this systematic review will be published in a peer-reviewed journal.

Trial registration number CRD42020162802

Keywords: Lung cancer; lung cancer screening; barriers; facilitators; systematic review

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Article Summary

Strengths and limitations of this study

(1) The mixed-methods systematic review will answer complementary research questions about the facilitators of, and barriers to uptake lung cancer practice at the level of participants and healthcare providers.

(2)The social-ecological model enables our analysis to illustrate the complex interplay of the interplay of individual, interpersonal, health system, and socio/environmental level influences on uptaking lung cancer screening.

(3) Our review will first analyze the similarity and difference of barriers and facilitators among individuals with different risk statuses and screening time, and adults and healthcare providers, which help to provide tailored LCS recommendations or interventions for specific individuals.

(4) This review will not include grey literature and studies published in languages other than English and Chinese and this will contribute to reporting bias.

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Introduction

Lung cancer is the second most commonly diagnosed cancer and the leading cause of cancer death. Worldwide, an estimated 2.2 million new lung cancer cases and 1.8 million deaths occurred in 2020, representing approximately one in 10 (11.4%) cancers diagnosed and one in 5 (18.0%) deaths.[1] International agency for cancer research reported that the incidence of lung cancer has been on the rise globally, especially in China.[2] China is accounted for the largest numbers of lung cancer patients in the world, it is expected that the number of lung cancer patients in China will reach 1million by 2025.[2,3] Due to the asymptomatic nature of lung cancer, they are often diagnosed at an advanced stage when the prognosis is poor or futile, with only 12%~16% survival rate at five years.[4,5] While the five-year survival rate increased to 60%~70% when diagnosed at an early stage.[5,6] This evidence highlighting the importance of secondary prevention for decreasing mortality of lung cancer and the improvement of a five-year survival rate.

At present, low-dose computed tomography (LDCT) has been demonstrated to be a sensitive tool for the detection of early-stage lung cancer.[7] Compared to chest X-ray (CXR), LDCT can significantly decrease the lung cancer mortality rate by 20% and all-cause mortality by 7%.[8] In 2013, the United States Preventive Services Task Force recommends (grade B) annual LDCT screening for high-risk individuals.[9] In 2018, a survey of 100,000 urban Chinese smokers aged 45-80 years found that there would be a lung cancer mortality reduction of 17.2% and 24.2% by LDCT screening when compared to CXR screening and no screening, respectively.[6] Therefore, LDCT may be effective in reducing lung cancer morbidity and mortality of lung cancer, and decreasing social burden.[10]

Despite the supportive landscape for lung cancer screening (LCS) with LDCT, globally, uptake rates of LCS with LDCT remain low.[2,5,11] The percentage of the eligible population who had received LCS only increased from 3.3% in 2010 to 3.9% in 2015 in the United States.[12] In China, from 2009 to 2012, the Cancer screening program has been launched successively in rural and urban areas, but the uptake rates of LDCT still low ranged from 6.4% to 31.91%.[13,14,15] Furthermore, the uptake of

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LCS is varied by the screening time and the risk status of participants.[2,16] Thus, understanding the barriers and facilitators to uptake of LCS practice is an important premise to promote the successful implementation of LCS programs and maximize the secondary prevention effect of LDCT.

Barriers and facilitators are the factors that inhibit and encourage a person from participating in LCS services. Many original studies relevant to the underuse of LCS suggest a range of barriers, related to individuals, healthcare professionals, health organizations, and system level.[2,5] Numerous factors included lack of awareness, lack of opportunity/access, individual beliefs, economic obstacles, et al, have been described as barriers to participation of LCS.[2,5,17,18] However, the participants, research settings, outcomes varied among these studies. For example, in terms of smoking history, Duong et al (2017)[19]found that current smokers showed less adherence to LCS, compared with patients who smoked less than they used to or quit smoking, which conflicted with the findings of Delmerico et al (2014)[20] that reported no difference among different smoking status groups. To draw well-informed conclusions of LCS practice that may be of use in shaping future public health efforts, it is necessary to conduct a systematic review of the current research on factors that may affect the uptake of LCS.

There have been several systematic reviews address LCS practice, however, most of the publications concerning the benefits, harms, and effectiveness of LDCT.[3,10,21,22] To the best of our knowledge, only three systematic reviews address barriers and facilitators to LCS uptake. Stefanie et al (2018)[23] reviewed 14 studies that evaluated the gender and social difference of participants in LCS programs, but the review evidences did not reflect the whole profile of factors influencing LCS practice. Because of the description of the socio-ecological theory, health behavior, such as LCS uptake behavior, is influenced by biological, psychological, socio-cultural, and environmental factors.[24] The other integrative review[5] is summarized 10 articles on the barriers to screening for lung cancer with LDCT, but is focused exclusively on the eligible U.S. population, and the quality of included articles was not assessed, which influencing the generalization of review evidence. And another systematic review is

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summarized 15 articles of high-risk patients' adherence to periodic LCS reported in cohort studies, but only included studies that were conducted in the US as well. Besides, the follow-up period was shorter than seen in the clinical trials and there was heterogeneity of the LCS eligibility criteria across the included studies, suggesting that future research should consider how differences in patients' risk of lung cancer impacts their adherence to screening.[25] Furthermore, the relevant reviews all explore the views of participants, not healthcare providers, who may also provide relevant insights into factors related to health systems and resources.[26]

Therefore, the aim of this mixed-methods systematic review is to identify, appraise and synthesize the available qualitative and quantitative literature to gain understanding as to: (1) what are the facilitators of, and barriers to uptake LCS practice at the level of participants and health-care providers? (2) what is the difference or similarities of barriers and facilitators to uptake LCS among subgroups(e.g., participants vs health-care providers, initial vs ongoing LCS behavior, high-risk, moderate-risk vs low-risk individuals)? (3)identify gaps in the literature to inform future implementation practice. Our findings can help to identify effective strategies that address barriers and leverage facilitators which may inform the development of LCS promotional and education programs to increase screen uptake among eligible adults, ultimately improve timely diagnosis for lung cancer and reduce lung cancer mortality.

Methods

We are using mixed methods methodology within this systematic review to answer complementary research questions within one study. This systematic review will be conducted following the Evidence for Policy and Practice Information and Co-ordinating Centre approach for mixed methods reviews,[27] and reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Protocol guidelines,[28] which has been registered with the International Prospective Register of Systematic Reviews (PROSPERO) (Registration number: CRD42020162802).

Search strategy

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To develop the searching strategy of this systematic review, we broke the review question into PICOs components(see Table 1). Boolean terms "OR" and "AND", Medical Subject Heading (MeSH), and truncation "*" will be used. The searching strategy is mainly composed of words expressing P, I, and O. The search strategy of PubMed is shown in Table 2, and the full search strategy is available in Appendix 1, which is reviewed and checked by one university librarian. The search strategies for selecting the fields of title, abstract, or keyword will be adjusted according to different characteristics of databases. Studies will be exported to EndNode X9 and duplicates will be deleted.

Search databases

We will perform an all-round search for published studies in 9 electronic databases from their inception up to December 31, 2020: PUBMED, Ovid (Journals @ Ovid Full Text and Ovid MEDLINE), EMBASE, CINAHL, PsycINFO, Cochrane Library, Chinese Biomedical Database (CBM), Chinese National Knowledge Infrastructure (CNKI), and Wanfang database. The database search was supplemented with searches on Google scholar, Proquest Theses and Dissertations database, manually examining reference lists of systematic review and included articles.

Eligibility criteria

Quantitative, qualitative or mixed-method studies published in English or Chinese language examining barriers and/or facilitators of uptake LCS practice will be included. In this study, uptake LCS practice includes the actual LCS behavior and intention or decision to LCS. Table 1 summarizes the elements of the inclusion and exclusion criteria based on the population, interest of study, comparison, outcome, and study design. Quantitative studies will be included to identify associations between various factors and screening behavior, while qualitative studies will be included to explore barriers and facilitators to screening that were reported by participants. Mixed methods studies whose quantitative or qualitative components meet the inclusion criteria will be included.

Study selection

According to the predefined eligibility criteria, two authors (YAL & YTH) will

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independently assess the titles and abstracts, and exclude the irrelevant articles. Then, all potential relevant full-text articles will be assessed for eligibility. Any disagreements arising will be resolved by discussion with a third reviewer (FFH). A Cohen's kappa coefficient will be calculated to assess the selection consistency between them. The search strategy and study selection process will be documented using a PRISMA flow diagram.

Data extraction

One primary reviewer will independently extract data from selected studies using structured and standardized data extraction forms (see Appendix 2) and later across checked by another reviewer. If any, we will try to contact the corresponding author to obtain the data we need. In instances where multiple publications relate to the same study, these will be reported together. The following domains will be retrieved: the authors' name, year of publication, country and setting, sample characteristics, study design, research focus, and findings. For quantitative studies, data will be extracted including the barriers and facilitators that were significantly associated with LCS practice, as well as proportions of participants reporting a barrier or facilitator. For qualitative studies, data will be extracted including all reported barriers/facilitators. Data from the studies will be used to form a narrative analysis of barriers and facilitators to LCS based on emergent themes.

Quality assessment

For eligible primary studies, quality assessment will be undertaken to illustrate potential sources of bias. As we anticipate the majority of eligible studies will be observational, studies will not automatically be excluded on the basis of 'low' quality assessment if they are considered to contribute relevant information. We propose using the NIH Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies,[29] the Critical Appraisal Skills Programme criteria adapted for qualitative studies for evaluating qualitative research,[30] and the 16-item quality assessment tool (QATSDD) for mixed-methods study.[31] Quality assessment of primary studies will be undertaken independently by two reviewers and recorded in an excel spreadsheet. An overall assessment of 'high', 'medium', 'low', or 'unclear' will be assigned and

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reported.

Data synthesis

For our first objective, the extracted data will be further summarized and classified using the social-ecological model from the five nested, hierarchical levels, that is, individual, interpersonal, health system, and socio/environmental factors (see Appendix 3). In terms of quantitative studies, we will provide a descriptive summary highlighting the most frequent themes within each level, and use frequency analysis to provide a numerical summary of the charted data. Records drawing from the same study dataset will be treated as one unit of analysis. In this study, if sufficient data were reported, we propose three subgroup analyses to compare the impact of LCS by 1) screening time (initial vs ongoing), 2) risk status (high-risk, moderate-risk, vs low-risk), and 3) people (participants vs healthcare providers).

For the qualitative studies, Thomas and Harden's thematic synthesis method[32] will be used, that is, we will pool the results from individual primary studies by initially separating the findings, coding and interpreting the text and then combining them through the identification of key themes across the studies as well as similarities and differences within those themes.

For our second objective, we will be overarching and comparing the differences or similarities explored within the domains from data synthesis mentioned above, between the participants and health-care providers, and adults with different screening time and risk status of lung cancer.

For our third objective, we will examine the themes in which there are none or few barriers and facilitators identified. The paucity of identified barriers and facilitators within these themes may represent areas that are not relevant for LCS uptake or where a gap in the literature may exist. We will discuss and provide examples of barriers and facilitators that were not captured in the literature. In addition, we will analyze the data on the study limitations or implications, ad described by authors, to characterize areas for further research.

Patient and public involvement

There was no patient or public involvement in the design or planning of the study.

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Ethics and dissemination

The current study is based on a review of published data and hence does not require ethical approval. Our protocol follows a rigorous methodology, using a theory-based approach that provides for a systematic understanding of the factors contributing to the underuse of LCS. As far as we are aware, this will be the first systematic review to collate comprehensive evidence in relation to barriers and facilitators to uptake of LCS programs based on the social-ecological model. We anticipate the results of this study may be of interest to national and international stakeholders interested in improving uptake in adult LCS programs.

Firstly, the findings from this review will illustrate the complex interplay of the interplay of individual, interpersonal, health system, and socio/environmental barriers that inhibit individuals from uptaking LCS and also highlight key enabling factors. Our findings can enable provide information about how LCS uptake may be improved and is of particular importance given that LCS uptake in the developed or developing countries is low but lung cancer is a pressing public health issue. Especially in China, although in recent years, municipal or city-level screening programs are increasing since the launch of Rural China screening program in 2009 and Cancer Screening Program in Urban China in 2012, there is still a desperate lack of pragmatic clinical trials on LCS and poor uptake rate.[2]

Secondly, this review focuses on the views of adults as well as healthcare providers which thus contributes valuable information regarding the perspective of the target group for screening, as well as insights from professionals who provide this service. The collective evidence may guide the development of health promotion programs that incorporate the views of the target group.

Thirdly, the similarity and difference of barriers and facilitators among individuals with different risk status and screening time, and adults and healthcare providers, are summarized in this review will help to provide tailor LCS recommendations or interventions for specific individuals.

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Author contributions: All authors have read and approved the final manuscript. FFH: conception and design of the study, drafting of the manuscript; HMX: participated in the design of the study, refinement of the literature search strategy and drafting of the protocol; YAL: assisted with the review of the literature and drafting of the initial manuscript; YTH and BNC: design of the study and formulation of the data extraction procedure for outcomes of interest.

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Competing interests statement: None declared.

Patient consent for publication: Not required.

Data sharing statement: Technical appendix, statistical code, and dataset available from the Dryad repository.

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Table 1 Eligibility criteria for studies to be included in the systematic review.

PICOS item	Inclusion criteria	Exclusion criteria
P-population	Studies involving lung cancer screened or unscreened adults or healthcare providers involved in LCS.	Previously received a diagnosis of lung cancer or any other cancer.
I-interest of research	(1) Studies describing barriers or facilitators to uptake LCS at the individuals or level, including the actual LCS behavior and intention or decision to LCS (2) Studies describing barriers or facilitators to uptake LCS at healthcare providers level.	Studies with no research method examining barriers or facilitators of LCS.
C-comparison	Not applicable, if a paper also reported on the differences between people with and people without screening as part of the study then this information was synthesised as part of the analysis.	
O-outcome(s)	(1)Barriers to uptake of and adherence to LCS. (including the actual LCS behavior and intention or decision to LCS) (2)Enablers of uptake of and adherence to LCS (including the actual LCS behavior and intention or decision to LCS) (3)Rates of uptake of and adherence to LCS(including the actual LCS behavior and intention or decision to LCS)	Studies that do not describe at least one of: barriers, enablers or determinants of uptake/adherence of LCS.
S-study design	Quantitative, qualitative or mixed-methods studies	(1) Conference abstracts, reviews, editorials, opinion pieces, dissertations, letters, books and full-text would not present or available original data even on requesting from the author. (2) Unpublished manuscripts and conference abstracts (3)Duplicate studies: for studies published with the same or different titles or in more than one journal, the most updated version shall be considered.

PICOS, population, intervention, comparison, outcome and study; LCS, lung cancer screening.

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Table 2 Search strategy of PubMed.

Search items	
1	(Small Cell Lung Carcinoma) OR (Carcinoma, Non-Small-Cell Lung)
2	Lung cancer OR NSLC OR NSCLC OR SLC OR SCLC
3	Lung neoplasm*
4	1 OR 2 NOT 3
5	(low dose computed tomography) OR (low dose CT) OR LDCT
6	scan* or screen*
7	5 AND 6
8	(lung cancer screen*) OR LCS
9	(Early Detection of Cancer)
10	7 OR 8 OR 9
11	Knowledge OR (Health Knowledge, Attitudes, Practice) OR Awareness OR Attitude OR (Attitude to Health) OR Behaviour OR (Patient Compliance) OR Perception
12	barrier* OR facilitator* OR utilisation* OR awareness OR knowledge OR attitudes OR practice OR belief OR adherence OR compliance OR perception OR perspective
13	11 OR 12
14	4 AND 10 AND 13

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Appendix 1 the full search strategy of PubMed

(((((((barrier*[Title/Abstract] OR facilitator*[Title/Abstract] OR utilisation*[Title/Abstract] OR awareness[Title/Abstract] OR knowledge[Title/Abstract] OR attitudes[Title/Abstract] OR practice[Title/Abstract] OR belief[Title/Abstract] OR adherence[Title/Abstract] OR compliance[Title/Abstract] OR perception[Title/Abstract] OR perspective[Title/Abstract])) AND Humans[Mesh])) OR ((((((Behavior[MeSH Terms]) OR ((Attitude[MeSH Terms]) OR Attitude to Health[MeSH Terms])) OR Awareness[MeSH Terms]) OR ((Knowledge[MeSH Terms]) OR Health Knowledge, Attitudes, Practice[MeSH Terms])) OR Patient Compliance[MeSH Terms]) OR Perception[MeSH Terms])) AND Humans[Mesh])) AND (((((((((scan*[Title/Abstract]) OR screen*[Title/Abstract])) AND (("low dose computed tomography"[Title/Abstract]) OR low dose CT[Title/Abstract]) OR LDCT[Title/Abstract]))) OR ((lung cancer screen*[Title/Abstract]) OR LCS[Title/Abstract]))) OR Early Detection of Cancer[MeSH Terms]) AND Humans[Mesh])) AND ((((((Small Cell Lung Carcinoma[MeSH Terms]) OR Carcinoma, Non-Small-Cell Lung[MeSH Terms])) OR (((((lung cancer[Title/Abstract]) OR SCLC[Title/Abstract]) OR SLC[Title/Abstract]) OR NSCLC[Title/Abstract]) OR NSLC[Title/Abstract]))) NOT lung neoplasm*[Title/Abstract]) Filters: Humans

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Appendix 2 Structured and standardized data extraction forms

Data items	Description
Reference ID number	ID number in citation management software
Author(s), year of publication	First author, article year
Geographic location	In which country/city was the study conducted
Study setting	Where did the study take place (including context).
Study design	The study design as defined by authors.
Research purpose/ focus	The aims of the study as defined by the author.
Population	Number, types and characteristics of participants of the study, including screening time and risk status. (sampling methods, sampling numbers, age (range & mean), gender(%), smoking status, screening individuals or healthcare providers, screening time and risk status(n(%)))
Outcomes	How the authors measured outcomes and the main results. Intention to screen: n(%)
Barriers	Factors that may have reduced uptake of LCS at the level of individuals and healthcare providers.
Facilitators	Factors that may have enabled uptake of LCS at the level of individuals and healthcare providers.
Theoretical basis	If applicable, theories and frameworks described in the study for the categorization of barriers and facilitators.
Study limitations	Authors' reported gaps and limitations of the study.
Study implications	Authors' reported future research direction.

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Appendix 3 Data synthesize forms

Data items	**Social-ecological model**	**Subgroup analysis**			**Outcome description**			
Barriers /facilitators	Including individual, interpersonal, health system, and socio/environmental factors	Screening time	Risk status	Individuals	Identified in a quantitative study (# studies & statistically significant results or proportion)	Identified in a qualitative study (# studies & tresults)	Identified in a Mixed methods study (# studies& results)	Total # studies in which identified

Note: screening time (initial vs ongoing), risk status (high-risk, moderate-risk, vs low-risk), and individuals (participants vs healthcare providers).

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	10

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	10
Sponsor	#5b	Provide name for the review funder and / or sponsor	10
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	10
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7(Table 1)
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7(Table 2、Appendix 1)
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-8

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Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7-8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8(Appendix 2)
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7(Table 1)
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7(Table 1)
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A
Confidence in cumulative	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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evidence

None The PRISMA-P elaboration and explanation paper is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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Barriers and facilitators of lung cancer screening uptake: Protocol of a mixed methods systematic review

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Barriers and facilitators of lung cancer screening uptake: protocol of a mixed methods systematic review

Yu-an Lin[1], Yu ting Hong[1],Bo ni Chen[1], Hui ming Xiao[1] , Fei fei Huang[1]

1. School of Nursing, Fujian Medical University, Fuzhou, China

Email：

Yu-an Lin: lya980211@163.com

Yu ting Hong: ann_hyt@163.com

Bo ni Chen: chenboni2005@163.com

Hui ming Xiao: huimin_xiao@126.com

Correspondence to Fei fei Huang:

hff315@fjmu.edu.cn

No 1 Xueyu Road, Minhou county, Fuzhou 350108, Fujian, China.

Telephone number +86 13763869099

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Abstract

Introduction: The global uptake rates of lung cancer screening (LCS) with low-dose computed tomography remain low. Since numerous factors contribute to the underuse of LCS, a theory-informed approach to identify and address the uptake of LCS barriers and facilitators is required. This study aims to document the methods which were used to identify, appraise, and synthesise the available qualitative, quantitative, and mixed methods evidence, addressing the barriers and facilitators at the individual and healthcare provider level, according to the social-ecological model, before identifying gaps to aid future practices and policies.

Methods and analysis: The following databases were searched: PubMed, Ovid (Journals @ Ovid Full Text and Ovid MEDLINE), EMBASE, CINAHL, PsycINFO, Cochrane Library, Chinese Biomedical Database (CBM), Chinese National Knowledge Infrastructure (CNKI), and Wanfang database, from their creation up to 31 December 2020. Two reviewers were involved in independently screening, reviewing, and synthesising the data; and calibration exercises were conducted at each stage. Disagreements between the two authors were resolved by arbitration by a third author. The Critical Appraisal Checklist for Studies Reporting Prevalence Data from the Joanna Briggs Institute, the Critical Appraisal Skills Programme criteria adapted for qualitative studies, and the 16-item Quality Assessment Tool (QATSDD) were used in the quality assessment of primary studies. We performed data synthesis using the Review Manager (RevMan) software, version 5.3.

Ethics and dissemination: This study is a review of published data and therefore needs no ethical approval. The findings of this systematic review will be published in a peer-reviewed journal.

Trial registration number: CRD42020162802.

Keywords: Lung cancer; lung cancer screening; barriers; facilitators; systematic review.

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Article Summary

Strengths and limitations of this study

(1) The mixed-methods systematic review answers complementary research questions about the facilitators and barriers of lung cancer uptake practices at the level of participants and healthcare providers.

(2)The social-ecological model enables our analysis to illustrate the complex interplay of the individual, interpersonal, health system, and socio/environmental aspects influencing the lung cancer screening uptake.

(3) Our review first analyses similarities and differences between barriers and facilitators among individuals with different risk statuses and screening times, and between adults and healthcare providers, which helps to provide tailored LCS recommendations or interventions for specific individuals.

(4) This review does not include grey literature and studies published in languages other than English and Chinese, which leads to reporting bias.

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1. Introduction

Lung cancer is the second most commonly diagnosed type of cancer, as well as the leading cause of cancer-related deaths. Worldwide, an estimated 2.2 million new lung cancer cases and 1.8 million deaths occurred in 2020, representing approximately one in 10 (11.4%) cancers diagnosed and one in 5 (18.0%) deaths.[1] The International Agency for Cancer Research reported that the incidence of lung cancer has been increasing globally, especially in China.[2] The country accounts for the largest number of lung cancer patients worldwide and is expected to reach 1 million by 2025.[2,3] Due to the asymptomatic nature of lung cancer, patients are often diagnosed at an advanced stage when the prognosis is poor or futile, with only 12%–16% survival rate at five years.[4,5] The same five-year survival rate increases to 60%–70% when the disease is diagnosed at an early stage.[5,6] This evidence highlights the importance of secondary prevention to decrease lung cancer mortality and improve the five-year survival rate.

Low-dose computed tomography (LDCT) has been proven as a sensitive tool for the detection of early-stage lung cancer.[7] Compared to chest X-ray (CXR), LDCT can significantly decrease the lung cancer mortality rate by 20% and all-cause mortality by 7%.[8] In 2013, the United States Preventive Services Task Force recommended (grade B) annual LDCT screening for high-risk individuals.[9] In 2018, a survey of 100,000 urban Chinese smokers aged 45–80 years found that LDCT screening would reduce lung cancer mortality by 17.2% and 24.2%, compared to CXR screening and no screening, respectively.[6] Therefore, LDCT may be effective in reducing lung cancer morbidity and mortality rates, alleviating social burden.[10]

Despite the overall support for lung cancer screening (LCS) with LDCT, its uptake rates remain low.[2,5,11] In the US, the percentage of eligible individuals who have received LCS increased only slightly from 3.3% in 2010 to 3.9% in 2015.[12] In China, from 2009 to 2012, the Cancer Screening Program has been launched successively in rural and urban areas, but the uptake rates of LDCT are still low, ranging from 6.4% to 31.91%.[13,14,15] Furthermore, the uptake of LCS varies according to the screening time and the risk status of the participants.[2,16] Thus,

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understanding the barriers and facilitators of the LCS uptake practice is an important premise to promote the successful implementation of LCS programs and maximise the secondary prevention effect of LDCT.

Barriers and facilitators are factors that inhibit or encourage an individual to participate in LCS services. Many original studies relevant to the underuse of LCS suggest a range of barriers related to individuals, healthcare professionals, health organisations, and the system level.[2,5] Numerous factors, including the lack of awareness, the lack of opportunity/access, individual beliefs, economic obstacles, and others, have been described as barriers to participation in LCS.[2, 5,17,18] However, the participants, research settings, and outcomes of those studies varied. For example, regarding smoking history, Duong et al. (2017) found that current smokers showed less adherence to LCS than patients who smoked less than they used to or quit smoking,[19] which conflicted with the findings of Delmerico et al. (2014),[20] who reported no difference among various groups with smoking status. To draw well-informed conclusions of LCS practices that would aid the shaping of future public health efforts, it is necessary to conduct a systematic review of the current research on factors affecting the LCS uptake.

There have been several systematic reviews addressing the LCS practice; however, most publications examine the benefits, harms, and effectiveness of LDCT. [3,10,21,22] To the best of our knowledge, only three systematic reviews address the barriers and facilitators of the LCS uptake. Stefanie et al. (2018) reviewed 14 studies [23], evaluating the gender and social differences of participants in LCS programs, but the review evidence did not analyse the factors influencing the LCS practice. According to the socio-ecological theory, health behaviour, such as LCS uptake behaviour, is influenced by biological, psychological, sociocultural, and environmental factors.[24] The second integrative review [5] summarized 10 articles on the barriers of LDCT lung cancer screening but focused exclusively on the eligible US population, and the quality of the articles included was not assessed, which influenced the generalisation of review evidence. The third systematic review summarised 15 articles on high-risk patients' adherence to periodic LCS reported in

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cohort studies but included only those studies that were conducted in the US. In addition, the follow-up period was shorter than that seen in the clinical trials, and there was heterogeneity of the LCS eligibility criteria across the included studies, suggesting that future research should consider how differences in patients' risk of lung cancer impact their adherence to screening.[25] Furthermore, the relevant reviews explore the views of participants, not healthcare providers, who may also provide relevant insights into the factors related to health systems and resources.[26]

Therefore, the aim of this mixed-methods systematic review is to identify, appraise, and synthesise the available qualitative and quantitative literature to improve our understanding of: (1) What are the facilitators of and barriers to LCS practice uptake at the level of participants and healthcare providers? (2) What are the differences or similarities of barriers and facilitators to LCS uptake among subgroups (e.g. participants vs. healthcare providers, initial vs. ongoing LCS behaviour, high-risk vs. moderate-risk vs. low-risk individuals)? (3) What are some remaining gaps in the literature which we can examine to inform and aid future implementation practices? Our findings can help identify effective strategies regarding barriers and leverage facilitators which may aid the development of LCS promotional and education programs to increase screen uptake among eligible adults, ultimately improving timely lung cancer diagnosis and reducing the disease's mortality.

2. Methods and analysis

This systematic review adopted a mixed methods methodology to answer its complementary research questions. The review was conducted following the Evidence for Policy and Practice Information and Co-ordinating Centre approach for mixed methods reviews[27] and reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Protocol guidelines[28] which have been registered with the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD42020162802).

Search strategy

To develop the search strategy for this systematic review, we broke down the review question into PICO components (see Table 1). Boolean terms 'OR' and 'AND',

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Medical Subject Heading (MeSH), and truncation '*' was used. The search strategy was mainly composed of words expressing P, I, and O. The search strategy of PubMed is shown in Table 2, and the full search strategy is available in Appendix 1, which was reviewed and edited by a university librarian. The search strategies for selecting titles, abstracts, or keywords were adjusted according to different database characteristics. Studies were exported to EndNode X9, and duplicates were deleted.

Search databases

We performed an all-round search for published studies in nine electronic databases, from their creation up to 31 December 2020: PubMed, Ovid (Journals @ Ovid Full Text and Ovid MEDLINE), EMBASE, CINAHL, PsycINFO, Cochrane Library, Chinese Biomedical Database (CBM), Chinese National Knowledge Infrastructure (CNKI), and Wanfang databases. The database search was supplemented with Google Scholar searches, manually examining reference lists of systematic reviews and articles.

Eligibility criteria

Quantitative, qualitative, or mixed-method studies published in English or Chinese examining the barriers and/or facilitators of LCS uptake practice were included. In this study, the LCS uptake practice included LCS behaviours and the intention or decisions about LCS. Table 1 summarises the elements of the inclusion and exclusion criteria based on population, study interest, comparison, outcome, and study design. Quantitative studies were included to identify associations between various factors and screening behaviour, while qualitative studies were used to explore screening barriers and facilitators reported by participants. Mixed methods studies whose quantitative or qualitative components meet the inclusion criteria were included.

Study selection

According to the predefined eligibility criteria, two authors (YAL & YTH) with experience in conducting systematic reviews independently assessed the titles and abstracts and excluded irrelevant articles. Then, all potentially relevant full-text articles were assessed for eligibility. Any disagreements arising were resolved by a discussion with a third reviewer (FFH), who also has experience with systematic

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reviews. Cohen's kappa coefficient was calculated to assess selection consistency between them. The search strategy and study selection process were documented using a PRISMA flow diagram.

Data extraction

One primary reviewer independently extracted data from selected studies using structured and standardised data extraction forms (see Appendix 2), which were then cross-checked by another reviewer. If any, we attempted to contact the corresponding author to obtain the required data. When multiple publications were related to the same study, these were reported together. The following domains were retrieved: author name, year of publication, country and setting, sample characteristics, study design, research focus, and findings. For the study findings, the barriers and facilitators associated with LCS practice were extracted and mapped onto the social-ecological model. The results of the quantitative study were recorded (e.g. r coefficient, β value, p-values). Data from the qualitative studies were used to form a narrative analysis of LCS barriers and facilitators based on emergent themes.

Quality assessment

Quality assessment was undertaken for eligible primary studies to illustrate the potential sources of bias. As we anticipated that the majority of eligible studies would be observational, studies were not automatically excluded on the basis of 'low' quality assessment if they were considered to contribute with relevant information. We proposed using the Critical Appraisal Checklist for Studies Reporting Prevalence Data from the Joanna Briggs Institute,[29] the Critical Appraisal Skills Programme criteria adapted for qualitative studies for evaluating qualitative research,[30] and the 16-item Quality Assessment Tool (QATSDD) for a mixed methods study.[31] Quality assessment of the primary studies was undertaken independently by two reviewers and recorded in an Excel spreadsheet. An overall assessment of 'high', 'medium', 'low', or 'unclear' was assigned and reported.

Data synthesis

For our first objective, the extracted data were further summarised and classified using the social-ecological model from the five nested, hierarchical levels: individual,

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interpersonal, health system, and socio/environmental factors (see Appendix 3).

Regarding quantitative studies, heterogeneity was assessed by a visual inspection of the forest plot and chi-squared test and then quantified using the I^2 statistic before the pooling of studies.[32,33] In the event of statistically significant heterogeneity, we reported our results as a narrative. Heterogeneity was further explored using subgroup analysis. If sufficient data were reported in this study, we proposed three subgroup analyses, comparing the impact of LCS by 1) screening time (initial vs. ongoing), 2) risk status (high-risk vs. moderate-risk, vs. low-risk), according to the criteria from Lung Cancer Screening, Version 3,[16] and 3) social groups (participants vs. healthcare providers). In the case of homogeneity, a meta-analysis was performed.

Regarding the qualitative studies, Thomas and Harden's thematic synthesis method[34] was adopted. That is, we pooled results from individual primary studies by initially separating the findings, coding and interpreting the text, and then combining them through identifying the key themes across the studies, including similarities and differences within those themes.

Regarding our second objective, we were overarching and comparing the differences and similarities explored within the domains from the data synthesis mentioned above, between the participants and healthcare providers, and adults with different screening times and lung cancer risk status.

Regarding our third objective, we examined the themes with no or few barriers and facilitators identified. The paucity of identified barriers and facilitators within these themes may represent areas irrelevant for LCS uptake or where a gap in the literature may exist. We discussed them and provided examples of barriers and facilitators that have not been captured in previous research. In addition, we analysed the data on the study limitations or implications described by the authors to distinguish areas for further research.

Patient and public involvement

There was no patient or public involvement in the design or planning of the study.

3. Ethics and dissemination

The current study was based on a review of published data and therefore required no

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ethical approval. Our protocol followed a rigorous methodology, using a theory-based approach providing a systematic understanding of the factors contributing to the underuse of LCS. To the best of our knowledge, this was the first systematic review to collate comprehensive evidence relating to barriers and facilitators of LCS programs uptake based on the social-ecological model. We anticipate that the results of this study may be of interest to national and international stakeholders interested in improving uptake in adult LCS programs.

First, the findings of this review illustrate the complex interplay of the individual, interpersonal, health system, and socio-environmental barriers that inhibit individuals from LCS uptake, highlighting key enabling factors. Our findings provide information on how LCS uptake can be improved and are particularly important since LCS uptake in developed or developing countries is low, though lung cancer remains to be a pressing public health issue. In China, although municipal or city-level screening programs have increased since the launch of the Rural China Screening Program in 2009 and the Cancer Screening Program in Urban China in 2012, there is still a desperate lack of pragmatic clinical trials (referring to proving the effectiveness of LCS or evaluating different implementation strategies to increase the uptake of LCS in eligible patients) on LCS and its low uptake rate.[2]

Second, this review focuses on the views of both adults and healthcare providers, which provide valuable information from the perspective of both the target group for screening and the professionals providing this service. The collective evidence may guide the development of health promotion programs incorporating the views of the target group.

Third, the similarities and differences between barriers and facilitators among individuals with different risk statuses and screening time, and adults and healthcare providers, are summarised in this review to help provide tailored LCS recommendations or interventions for specific individuals. Furthermore, this review also provides evidence about the associations and effects of LCS barriers and facilitators perceived by individuals and healthcare providers and about how healthcare is organised and financed transcend culture and geography.

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Author contributions: All authors on this paper meet the four criteria for authorship as identified by the International Committee of Medical Journal Editors (ICMJE); all authors have read and approved the final manuscript. FFH: conception and design of the study, drafting of the manuscript; HMX: participated in the design of the study, refinement of the literature search strategy, and drafting of the protocol; YAL: assisted with the review of the literature and drafting of the initial manuscript; YTH and BNC: design of the study and formulation of the data extraction procedure for outcomes of interest.

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Competing interests statement: None declared.

Patient consent for publication: Not required.

Data sharing statement: Technical appendix, statistical code, and dataset available from the Dryad repository.

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Tables

Table 1. Eligibility criteria for studies to be included in the systematic review.

PICOS item	Inclusion criteria	Exclusion criteria
P-population	Studies involving lung cancer screened or unscreened adults or healthcare providers involved in LCS.	Previously received a diagnosis of lung cancer or any other cancer.
I-interest of research	(1) Studies describing barriers or facilitators to LCS uptake at the individuals level, including the actual LCS behavior and intention or decision to LCS; (2) Studies describing barriers or facilitators to uptake LCS at healthcare providers level.	Studies with no research method examining LCS barriers or facilitators.
C-comparison	Not applicable, if a paper also reported on the differences between individuals with and without screening as part of the study then this information was synthesised as part of the analysis.	
O-outcome(s)	(1) Barriers to uptake of and adherence to LCS. (including the actual LCS behavior and intention or decision to LCS) (2) Enablers of uptake of and adherence to LCS (including the actual LCS behavior and intention or decision to LCS) (3)Rates of uptake of and adherence to LCS(including the actual LCS behavior and intention or decision to LCS)	Studies that do not describe at least one of: barriers, enablers or determinants of uptake/adherence of LCS.

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S-study design	Quantitative, qualitative or mixed-methods studies	(1) Conference abstracts, reviews, editorials, opinion pieces, dissertations, letters, books and full-text would not be presented or the original data were made available even on requesting from the author. (2) Unpublished manuscripts and conference abstracts (3)Duplicate studies: for studies published with the same or different titles or in more than one journal, the most updated version shall be considered.

PICOS, population, intervention, comparison, outcome and study; LCS, lung cancer screening.

Table 2. Search strategy of PubMed.

Search items	
1	(Small Cell Lung Carcinoma) OR (Carcinoma, Non-Small-Cell Lung)
2	Lung cancer OR NSLC OR NSCLC OR SLC OR SCLC
3	Lung neoplasm*
4	1 OR 2 NOT 3
5	(low dose computed tomography) OR (low dose CT) OR LDCT
6	scan* or screen*
7	5 AND 6
8	(lung cancer screen*) OR LCS
9	(Early Detection of Cancer)
10	7 OR 8 OR 9
11	Knowledge OR (Health Knowledge, Attitudes, Practice) OR Awareness OR Attitude OR (Attitude to Health) OR Behaviour OR (Patient Compliance) OR Perception
12	barrier* OR facilitator* OR utilisation* OR awareness OR knowledge OR attitudes OR practice OR belief OR adherence OR compliance OR perception OR perspective
13	11 OR 12
14	4 AND 10 AND 13

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Appendix 1 the full search strategy of each database

Database	Search strategy
PubMed	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. (lung cancer or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge"[Mesh] OR "Health Knowledge, Attitudes, Practice"[Mesh]) OR "Awareness"[Mesh]) OR ("Attitude"[Mesh] OR "Attitude to Health"[Mesh])) OR "Behavior"[Mesh]) OR "Patient Compliance"[Mesh]) OR "Perception"[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitudes or practice or belief or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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EMBASE	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab/Keyword] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab/Keyword] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
PsycINFO	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh]

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	10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
Ovid (Journals @ Ovid Full Text and Ovid MEDLINE)	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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CINAHL	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab/MM] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[MM] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[MM] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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Cochrane Library	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. (lung cancer or NSLC or NSCLC or SLC or SCLC) [ti/ab/kw] 3#. (lung neoplasm*)[ti/ab/kw] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT) [ti/ab/kw] 6# scan* or screen*[ti/ab/kw] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab/kw] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitudes or practice or belief or adherence or compliance or perception or perspective)[ti/ab/kw] 13#. 11 or 12 14#. 4 and 10 and 13
Chinese Biomedical Database (CBM)	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[ti/ab]

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	10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/ab] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
Chinese National Knowledge Infrastructure (CNKI)	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/kw/ab] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/kw/ab] 3#. (lung neoplasm*)[ti/kw/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/kw/ab] 6# scan* or screen*[ti/kw/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/kw/ab] 9#. "Early Detection of Cancer"[ti/kw/ab] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/kw/ab] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/kw/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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Chinese Wanfang Database	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/kw] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/kw] 3#. (lung neoplasm*)[ti/kw] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/kw] 6# scan* or screen*[ti/kw] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/kw] 9#. "Early Detection of Cancer"[ti/kw] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/kw] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/kw] 13#. 11 or 12 14#. 4 and 10 and 13

Note: MM: MeSH; ti: title; ab: abstract; kw: keyword

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Appendix 2 Structured and standardized data extraction forms

Data items	Description
Reference ID number	ID number in citation management software
Author(s), year of publication	First author, article year
Geographic location	In which country/city was the study conducted
Study setting	Where did the study take place (including context).
Study design	The study design as defined by authors.
Research purpose/ focus	The aims of the study as defined by the author.
Population	Number, types and characteristics of participants of the study, including screening time and risk status. (sampling methods, sampling numbers, age (range & mean), gender(%), smoking status, screening individuals or healthcare providers, screening time and risk status(n(%))
Outcomes	How the authors measured outcomes and the main results. Intention to screen: n(%)
Barriers	Factors that may have reduced uptake of LCS at the level of individuals and healthcare providers.
Facilitators	Factors that may have enabled uptake of LCS at the level of individuals and healthcare providers.
Theoretical basis	If applicable, theories and frameworks described in the study for the categorization of barriers and facilitators.
Study limitations	Authors' reported gaps and limitations of the study.
Study implications	Authors' reported future research direction.

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Appendix 3 Data synthesize forms

Data items	**Social-ecological model**	**Subgroup analysis**			**Outcome description**			
Barriers /facilitators	Including individual, interpersonal, health system, and socio/environmental factors	Screening time	Risk status	Individuals	Identified in a quantitative study (# studies & results)	Identified in a qualitative study (# studies & results)	Identified in a Mixed methods study (# studies& results)	Total # studies in which identified

Note: screening time (initial vs ongoing), risk status (high-risk, moderate-risk, vs low-risk), and individuals (participants vs healthcare providers).

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	10

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	10
Sponsor	#5b	Provide name for the review funder and / or sponsor	10
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	10
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7(Table 1)
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7(Table 2、Appendix 1)
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-8

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Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7-8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8(Appendix 2)
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7(Table 1)
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7(Table 1)
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A
Confidence in cumulative	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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evidence

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Barriers and facilitators of lung cancer screening uptake: Protocol of a mixed methods systematic review

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Barriers and facilitators of lung cancer screening uptake: protocol of a mixed methods systematic review

Yu-an Lin[1], Yu ting Hong[1],Bo ni Chen[1], Hui ming Xiao[1] , Fei fei Huang[1]

1. School of Nursing, Fujian Medical University, Fuzhou, China

Email：

Yu-an Lin: lya980211@163.com

Yu ting Hong: ann_hyt@163.com

Bo ni Chen: chenboni2005@163.com

Hui ming Xiao: huimin_xiao@126.com

Correspondence to Fei fei Huang:

hff315@fjmu.edu.cn

No 1 Xueyu Road, Minhou county, Fuzhou 350108, Fujian, China.

Telephone number +86 13763869099

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Abstract

Introduction: The global uptake rates of lung cancer screening (LCS) with low-dose computed tomography remain low. Since numerous factors contribute to the underuse of LCS, a theory-informed approach to identify and address the uptake of LCS barriers and facilitators is required. This study aims to document the methods which were used to identify, appraise, and synthesise the available qualitative, quantitative, and mixed methods evidence, addressing the barriers and facilitators at the individual and healthcare provider level, according to the social-ecological model, before identifying gaps to aid future practices and policies.

Methods and analysis: The following databases will be searched: PubMed, Ovid (Journals @ Ovid Full Text and Ovid MEDLINE), EMBASE, CINAHL, PsycINFO, Cochrane Library, Chinese Biomedical Database (CBM), Chinese National Knowledge Infrastructure (CNKI), and Wanfang database, from their creation up to 31 December 2020. Two reviewers will be involved in independently screening, reviewing, and synthesising the data; and calibration exercises will be conducted at each stage. Disagreements between the two authors will be resolved by arbitration by a third author. The Critical Appraisal Checklist for Studies Reporting Prevalence Data from the Joanna Briggs Institute, the Critical Appraisal Skills Programme criteria adapted for qualitative studies, and the 16-item Quality Assessment Tool (QATSDD) will be used in the quality assessment of primary studies. We will perform data synthesis using the Review Manager (RevMan) software, version 5.3.

Ethics and dissemination: This study is a review of published data and therefore needs no ethical approval. The findings of this systematic review will be published in a peer-reviewed journal.

Trial registration number: CRD42020162802.

Keywords: Lung cancer; lung cancer screening; barriers; facilitators; systematic review.

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Article Summary

Strengths and limitations of this study

(1) The mixed-methods systematic review answers complementary research questions about the facilitators and barriers of lung cancer uptake practices at the level of participants and healthcare providers.

(2)The social-ecological model enables our analysis to illustrate the complex interplay of the individual, interpersonal, health system, and socio/environmental aspects influencing the lung cancer screening uptake.

(3) Our review first analyses similarities and differences between barriers and facilitators among individuals with different risk statuses and screening times, and between adults and healthcare providers, which helps to provide tailored LCS recommendations or interventions for specific individuals.

(4) This review does not include grey literature and studies published in languages other than English and Chinese, which leads to reporting bias.

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1. Introduction

Lung cancer is the second most commonly diagnosed type of cancer, as well as the leading cause of cancer-related deaths. Worldwide, an estimated 2.2 million new lung cancer cases and 1.8 million deaths occurred in 2020, representing approximately one in 10 (11.4%) cancers diagnosed and one in 5 (18.0%) deaths.[1] The International Agency for Cancer Research reported that the incidence of lung cancer has been increasing globally, especially in China.[2] The country accounts for the largest number of lung cancer patients worldwide and is expected to reach 1 million by 2025.[2,3] Due to the asymptomatic nature of lung cancer, patients are often diagnosed at an advanced stage when the prognosis is poor or futile, with only 12%–16% survival rate at five years.[4,5] The same five-year survival rate increases to 60%–70% when the disease is diagnosed at an early stage.[5,6] This evidence highlights the importance of secondary prevention to decrease lung cancer mortality and improve the five-year survival rate.

Low-dose computed tomography (LDCT) has been proven as a sensitive tool for the detection of early-stage lung cancer.[7] Compared to chest X-ray (CXR), LDCT can significantly decrease the lung cancer mortality rate by 20% and all-cause mortality by 7%.[8] In 2013, the United States Preventive Services Task Force recommended (grade B) annual LDCT screening for high-risk individuals.[9] In 2018, a survey of 100,000 urban Chinese smokers aged 45–80 years found that LDCT screening would reduce lung cancer mortality by 17.2% and 24.2%, compared to CXR screening and no screening, respectively.[6] Therefore, LDCT may be effective in reducing lung cancer morbidity and mortality rates, alleviating social burden.[10]

Despite the overall support for lung cancer screening (LCS) with LDCT, its uptake rates remain low.[2,5,11] In the US, the percentage of eligible individuals who have received LCS increased only slightly from 3.3% in 2010 to 3.9% in 2015.[12] In China, from 2009 to 2012, the Cancer Screening Program has been launched successively in rural and urban areas, but the uptake rates of LDCT are still low, ranging from 6.4% to 31.91%.[13,14,15] Furthermore, the uptake of LCS varies according to the screening time and the risk status of the participants.[2,16] Thus,

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understanding the barriers and facilitators of the LCS uptake practice is an important premise to promote the successful implementation of LCS programs and maximise the secondary prevention effect of LDCT.

Barriers and facilitators are factors that inhibit or encourage an individual to participate in LCS services. Many original studies relevant to the underuse of LCS suggest a range of barriers related to individuals, healthcare professionals, health organisations, and the system level.[2,5] Numerous factors, including the lack of awareness, the lack of opportunity/access, individual beliefs, economic obstacles, and others, have been described as barriers to participation in LCS.[2, 5,17,18] However, the participants, research settings, and outcomes of those studies varied. For example, regarding smoking history, Duong et al. (2017) found that current smokers showed less adherence to LCS than patients who smoked less than they used to or quit smoking,[19] which conflicted with the findings of Delmerico et al. (2014),[20] who reported no difference among various groups with smoking status. To draw well-informed conclusions of LCS practices that would aid the shaping of future public health efforts, it is necessary to conduct a systematic review of the current research on factors affecting the LCS uptake.

There have been several systematic reviews addressing the LCS practice; however, most publications examine the benefits, harms, and effectiveness of LDCT. [3,10,21,22] To the best of our knowledge, only three systematic reviews address the barriers and facilitators of the LCS uptake. Stefanie et al. (2018) reviewed 14 studies [23], evaluating the gender and social differences of participants in LCS programs, but the review evidence did not analyse the factors influencing the LCS practice. According to the socio-ecological theory, health behaviour, such as LCS uptake behaviour, is influenced by biological, psychological, sociocultural, and environmental factors.[24] The second integrative review [5] summarized 10 articles on the barriers of LDCT lung cancer screening but focused exclusively on the eligible US population, and the quality of the articles included was not assessed, which influenced the generalisation of review evidence. The third systematic review summarised 15 articles on high-risk patients' adherence to periodic LCS reported in

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cohort studies but included only those studies that were conducted in the US. In addition, the follow-up period was shorter than that seen in the clinical trials, and there was heterogeneity of the LCS eligibility criteria across the included studies, suggesting that future research should consider how differences in patients' risk of lung cancer impact their adherence to screening.[25] Furthermore, the relevant reviews explore the views of participants, not healthcare providers, who may also provide relevant insights into the factors related to health systems and resources.[26]

Therefore, the aim of this mixed-methods systematic review is to identify, appraise, and synthesise the available qualitative and quantitative literature to improve our understanding of: (1) What are the facilitators of and barriers to LCS practice uptake at the level of participants and healthcare providers? (2) What are the differences or similarities of barriers and facilitators to LCS uptake among subgroups (e.g. participants vs. healthcare providers, initial vs. ongoing LCS behaviour, high-risk vs. moderate-risk vs. low-risk individuals)? (3) What are some remaining gaps in the literature which we can examine to inform and aid future implementation practices? Our findings can help identify effective strategies regarding barriers and leverage facilitators which may aid the development of LCS promotional and education programs to increase screen uptake among eligible adults, ultimately improving timely lung cancer diagnosis and reducing the disease's mortality.

2. Methods and analysis

This systematic review will adopt a mixed methods methodology to answer its complementary research questions. The review will be conducted following the Evidence for Policy and Practice Information and Co-ordinating Centre approach for mixed methods reviews[27] and reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Protocol guidelines[28] which have been registered with the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD42020162802).

Search strategy

To develop the search strategy for this systematic review, we break down the review question into PICO components (see Table 1). Boolean terms 'OR' and 'AND',

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Medical Subject Heading (MeSH), and truncation '*' are used. The search strategy is mainly composed of words expressing P, I, and O. The search strategy of PubMed is shown in Table 2, and the full search strategy is available in Appendix 1, which has been reviewed and edited by a university librarian. The search strategies for selecting titles, abstracts, or keywords will be adjusted according to different database characteristics. Studies will be exported to EndNode X9, and duplicates will be deleted.

Search databases

We will perform an all-round search for published studies in nine electronic databases, from their creation up to 31 December 2020: PubMed, Ovid (Journals @ Ovid Full Text and Ovid MEDLINE), EMBASE, CINAHL, PsycINFO, Cochrane Library, Chinese Biomedical Database (CBM), Chinese National Knowledge Infrastructure (CNKI), and Wanfang databases. The database search will be supplemented with Google Scholar searches, manually examining reference lists of systematic reviews and articles.

Eligibility criteria

Quantitative, qualitative, or mixed-method studies published in English or Chinese examining the barriers and/or facilitators of LCS uptake practice will be included. In this study, the LCS uptake practice will include LCS behaviours and the intention or decisions about LCS. Table 1 summarises the elements of the inclusion and exclusion criteria based on population, study interest, comparison, outcome, and study design. Quantitative studies will be included to identify associations between various factors and screening behaviour, while qualitative studies will be used to explore screening barriers and facilitators reported by participants. Mixed methods studies whose quantitative or qualitative components meet the inclusion criteria will be included.

Study selection

According to the predefined eligibility criteria, two authors (YAL & YTH) with experience in conducting systematic reviews independently will assess the titles and abstracts and excluded irrelevant articles. Then, all potentially relevant full-text articles will be assessed for eligibility. Any disagreements arising will be resolved by

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a discussion with a third reviewer (FFH), who also has experience with systematic reviews. Cohen's kappa coefficient will be calculated to assess selection consistency between them. The search strategy and study selection process will be documented using a PRISMA flow diagram.

Data extraction

One primary reviewer will independently extract data from selected studies using structured and standardised data extraction forms (see Appendix 2), which will be then cross-checked by another reviewer. If any, we will attempt to contact the corresponding author to obtain the required data. When multiple publications are related to the same study, these will be reported together. The following domains will be retrieved: author name, year of publication, country and setting, sample characteristics, study design, research focus, and findings. For the study findings, the barriers and facilitators associated with LCS practice will be extracted and mapped onto the social-ecological model [29]. That is, the factors were assigned to one of the four general categories: individual, interpersonal, health system, and socio/environmental. The results of the quantitative study will be recorded (e.g. r coefficient, β value, p-values). Data from the qualitative studies will be used to form a narrative analysis of LCS barriers and facilitators based on emergent themes.

Quality assessment

Quality assessment will be undertaken for eligible primary studies to illustrate the potential sources of bias. As we anticipate that the majority of eligible studies will be observational, studies will not be automatically excluded on the basis of 'low' quality assessment if they are considered to contribute with relevant information. We propose using the Critical Appraisal Checklist for Studies Reporting Prevalence Data from the Joanna Briggs Institute,[30] the Critical Appraisal Skills Programme criteria adapted for qualitative studies for evaluating qualitative research,[31] and the 16-item Quality Assessment Tool (QATSDD) for a mixed methods study.[32] Quality assessment of the primary studies will be undertaken independently by two reviewers and recorded in an Excel spreadsheet. An overall assessment of 'high', 'medium', 'low', or 'unclear' will be assigned and reported.

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Data synthesis

For our first objective, the extracted data will be further summarised and classified using the social-ecological model from the five nested, hierarchical levels: individual, interpersonal, health system, and socio/environmental factors (see Appendix 3).

Regarding quantitative studies, heterogeneity will be assessed by a visual inspection of the forest plot, which will be used to present the effect sizes for each included study in aggregate and the combined effect sizes in summary, and chi-squared test as well as quantified using the I^2 statistic before the pooling of studies. If there is statistical homogeneity between the studies ($I^2<50\%$), a fixed-effect model will be used for meta-analysis. If there is statistical heterogeneity ($I^2\geq50\%$) between the studies without clinical heterogeneity, the random effect model will be used for analysis. In the event of statistically significant heterogeneity, we will report our results as a narrative. [33,34] Heterogeneity will be further explored using subgroup analysis. If sufficient data are reported in this study, we will propose three subgroup analyses, comparing the impact of LCS by 1) screening time (initial vs. ongoing), 2) risk status (high-risk vs. moderate-risk, vs. low-risk), according to the criteria from Lung Cancer Screening, Version 3,[16] and 3) social groups (participants vs. healthcare providers). In the case of homogeneity, a meta-analysis will be performed.

Regarding the qualitative studies, Thomas and Harden's thematic synthesis method[35] will be adopted. That is, we will pool results from individual primary studies by initially separating the findings, coding and interpreting the text, and then combining them through identifying the key themes across the studies, including similarities and differences within those themes.

Regarding our second objective, we will overarch and compare the differences and similarities explored within the domains from the data synthesis mentioned above, between the participants and healthcare providers, and adults with different screening times and lung cancer risk status.

Regarding our third objective, we will examine the themes with no or few barriers and facilitators identified. The paucity of identified barriers and facilitators within

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these themes may represent areas irrelevant for LCS uptake or where a gap in the literature may exist. We will discuss them and provided examples of barriers and facilitators that have not been captured in previous research. In addition, we will analyse the data on the study limitations or implications described by the authors to distinguish areas for further research.

Patient and public involvement

There is no patient or public involvement in the design or planning of the study.

3. Ethics and dissemination

The current study is based on a review of published data and therefore requires no ethical approval. Our protocol follows a rigorous methodology, using a theory-based approach providing a systematic understanding of the factors contributing to the underuse of LCS. To the best of our knowledge, this is the first systematic review to collate comprehensive evidence relating to barriers and facilitators of LCS programs uptake based on the social-ecological model. We anticipate that the results of this study may be of interest to national and international stakeholders interested in improving uptake in adult LCS programs.

First, the findings of this review illustrate the complex interplay of the individual, interpersonal, health system, and socio-environmental barriers that inhibit individuals from LCS uptake, highlighting key enabling factors. Our findings provide information on how LCS uptake can be improved and are particularly important since LCS uptake in developed or developing countries is low, though lung cancer remains to be a pressing public health issue. In China, although municipal or city-level screening programs have increased since the launch of the Rural China Screening Program in 2009 and the Cancer Screening Program in Urban China in 2012, there is still a desperate lack of pragmatic clinical trials (referring to proving the effectiveness of LCS or evaluating different implementation strategies to increase the uptake of LCS in eligible patients) on LCS and its low uptake rate.[2]

Second, this review focuses on the views of both adults and healthcare providers, which provide valuable information from the perspective of both the target group for screening and the professionals providing this service. The collective evidence may

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guide the development of health promotion programs incorporating the views of the target group.

Third, the similarities and differences between barriers and facilitators among individuals with different risk statuses and screening time, and adults and healthcare providers, are summarised in this review to help provide tailored LCS recommendations or interventions for specific individuals. Furthermore, this review also provides evidence about the associations and effects of LCS barriers and facilitators perceived by individuals and healthcare providers and about how healthcare is organised and financed transcend culture and geography. Specifically, we will search a large number of relevant information to understand the organizational form, culture and environment of health care in the country or region where the eligible studies were conducted. Besides, we will further explore the barriers and facilitators in different cultural backgrounds, classify and summarize them, so as to analysis their similarities and differences, and explore the underlying reasons.

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Patient consent for publication: Not required.

Data sharing statement: Technical appendix, statistical code, and dataset available from the Dryad repository.

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Table 1. Eligibility criteria for studies to be included in the systematic review.

PICOS item	Inclusion criteria	Exclusion criteria
P-population	Studies involving lung cancer screened or unscreened adults or healthcare providers involved in LCS.	Previously received a diagnosis of lung cancer or any other cancer.
I-interest of research	(1) Studies describing barriers or facilitators to LCS uptake at the individuals level, including the actual LCS behavior and intention or decision to LCS; (2) Studies describing barriers or facilitators to uptake LCS at healthcare providers level.	Studies with no research method examining LCS barriers or facilitators.
C-comparison	Not applicable, if a paper also reported on the differences between individuals with and without screening as part of the study then this information was synthesised as part of the analysis.	
O-outcome(s)	(1) Barriers to uptake of and adherence to LCS. (including the actual LCS behavior and intention or decision to LCS) (2) Enablers of uptake of and adherence to LCS (including the actual LCS behavior and intention or decision to LCS) (3)Rates of uptake of and adherence to LCS(including the actual LCS behavior and intention or decision to LCS)	Studies that do not describe at least one of: barriers, enablers or determinants of uptake/adherence of LCS.

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S-study design	Quantitative, qualitative or mixed-methods studies	(1) Conference abstracts, reviews, editorials, opinion pieces, dissertations, letters, books and full-text would not be presented or the original data were made available even on requesting from the author. (2) Unpublished manuscripts and conference abstracts (3)Duplicate studies: for studies published with the same or different titles or in more than one journal, the most updated version shall be considered.

PICOS, population, intervention, comparison, outcome and study; LCS, lung cancer screening.

Table 2. Search strategy of PubMed.

Search items	
1	(Small Cell Lung Carcinoma) OR (Carcinoma, Non-Small-Cell Lung)
2	Lung cancer OR NSLC OR NSCLC OR SLC OR SCLC
3	Lung neoplasm*
4	1 OR 2 NOT 3
5	(low dose computed tomography) OR (low dose CT) OR LDCT
6	scan* or screen*
7	5 AND 6
8	(lung cancer screen*) OR LCS
9	(Early Detection of Cancer)
10	7 OR 8 OR 9
11	Knowledge OR (Health Knowledge, Attitudes, Practice) OR Awareness OR Attitude OR (Attitude to Health) OR Behaviour OR (Patient Compliance) OR Perception
12	barrier* OR facilitator* OR utilisation* OR awareness OR knowledge OR attitudes OR practice OR belief OR adherence OR compliance OR perception OR perspective
13	11 OR 12
14	4 AND 10 AND 13

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Appendix 1 the full search strategy of each database

Database	Search strategy
PubMed	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. (lung cancer or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge"[Mesh] OR "Health Knowledge, Attitudes, Practice"[Mesh]) OR "Awareness"[Mesh]) OR ("Attitude"[Mesh] OR "Attitude to Health"[Mesh])) OR "Behavior"[Mesh]) OR "Patient Compliance"[Mesh]) OR "Perception"[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitudes or practice or belief or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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EMBASE	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab/Keyword] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab/Keyword] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
PsycINFO	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh]

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	10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
Ovid (Journals @ Ovid Full Text and Ovid MEDLINE)	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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CINAHL	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab/MM] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[MM] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[MM] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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Cochrane Library	1#. "Small Cell Lung Carcinoma"[Mesh] OR "Carcinoma, Non-Small-Cell Lung"[Mesh] 2#. (lung cancer or NSLC or NSCLC or SLC or SCLC) [ti/ab/kw] 3#. (lung neoplasm*)[ti/ab/kw] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT) [ti/ab/kw] 6# scan* or screen*[ti/ab/kw] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab/kw] 9#. "Early Detection of Cancer"[Mesh] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[Mesh] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitudes or practice or belief or adherence or compliance or perception or perspective)[ti/ab/kw] 13#. 11 or 12 14#. 4 and 10 and 13
Chinese Biomedical Database (CBM)	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/ab] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/ab] 3#. (lung neoplasm*)[ti/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/ab] 6# scan* or screen*[ti/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/ab] 9#. "Early Detection of Cancer"[ti/ab]

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	10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/ab] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/ab] 13#. 11 or 12 14#. 4 and 10 and 13
Chinese National Knowledge Infrastructure (CNKI)	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/kw/ab] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/kw/ab] 3#. (lung neoplasm*)[ti/kw/ab] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/kw/ab] 6# scan* or screen*[ti/kw/ab] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/kw/ab] 9#. "Early Detection of Cancer"[ti/kw/ab] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/kw/ab] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/kw/ab] 13#. 11 or 12 14#. 4 and 10 and 13

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Chinese Wanfang Database	1#. "Small Cell Lung Carcinoma" OR "Carcinoma, Non-Small-Cell Lung"[ti/kw] 2#. ("lung cancer" or NSLC or NSCLC or SLC or SCLC) [ti/kw] 3#. (lung neoplasm*)[ti/kw] 4#. 1 or 2 not 3 5#. ("low dose computed tomography" or low dose CT or LDCT)[ti/kw] 6# scan* or screen*[ti/kw] 7#. 5 and 6 8#. "lung cancer screen*" or LCS[ti/kw] 9#. "Early Detection of Cancer"[ti/kw] 10#. 7 or 8 or 9 11#. ("Knowledge" OR "Health Knowledge, Attitudes, Practice" OR "Awareness" OR "Attitude" OR "Attitude to Health" OR "Behavior" OR "Patient Compliance" OR "Perception")[ti/kw] 12#. (barrier* or facilitator* or utilisation* or awareness or knowledge or attitude* or practice or belief* or adherence or compliance or perception or perspective)[ti/kw] 13#. 11 or 12 14#. 4 and 10 and 13

Note: MM: MeSH; ti: title; ab: abstract; kw: keyword

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Appendix 2 Structured and standardized data extraction forms

Data items	Description
Reference ID number	ID number in citation management software
Author(s), year of publication	First author, article year
Geographic location	In which country/city was the study conducted
Study setting	Where did the study take place (including context).
Study design	The study design as defined by authors.
Research purpose/ focus	The aims of the study as defined by the author.
Population	Number, types and characteristics of participants of the study, including screening time and risk status. (sampling methods, sampling numbers, age (range & mean), gender(%), smoking status, screening individuals or healthcare providers, screening time and risk status(n(%))
Outcomes	How the authors measured outcomes and the main results. Intention to screen: n(%)
Barriers	Factors that may have reduced uptake of LCS at the level of individuals and healthcare providers.
Facilitators	Factors that may have enabled uptake of LCS at the level of individuals and healthcare providers.
Theoretical basis	If applicable, theories and frameworks described in the study for the categorization of barriers and facilitators.
Study limitations	Authors' reported gaps and limitations of the study.
Study implications	Authors' reported future research direction.

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Appendix 3 Data synthesize forms

Data items	**Social-ecological model**	**Subgroup analysis**			**Outcome description**			
Barriers /facilitators	Including individual, interpersonal, health system, and socio/environmental factors	Screening time	Risk status	Individuals	Identified in a quantitative study (# studies & results)	Identified in a qualitative study (# studies & results)	Identified in a Mixed methods study (# studies& results)	Total # studies in which identified

Note: screening time (initial vs ongoing), risk status (high-risk, moderate-risk, vs low-risk), and individuals (participants vs healthcare providers).

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	10

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	10
Sponsor	#5b	Provide name for the review funder and / or sponsor	10
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	10
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7(Table 1)
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7(Table 2、Appendix 1)
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-8

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Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7-8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8(Appendix 2)
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7(Table 1)
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7(Table 1)
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A
Confidence in cumulative	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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evidence

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